

HintonMarket: Hintonville Farmers Market Management System

1. Background

In the fall of 2025, the Hintonville Library successfully deployed a new Library Information and Borrowing System (HinLIBS). While the system met its quality objectives, the project experienced significant cost overruns. Software project quality rests on three pillars: cost, features, and time. When HinLIBS prioritized features and time, costs exceeded the initial budget, leaving the Hintonville City Council and Library Board searching for revenue solutions.

The City Council and Library Board found a creative solution: convert the library's underused parking lot into a farmers market. This would create a community gathering space and generate revenue to offset the HinLIBS cost overrun.

To manage this new venture efficiently, the Town of Hintonville has issued a Request for Proposal (RFP) to develop HintonMarket, a Farmers Market Management System. This system must support vendor account management, stall bookings, compliance verification, and waitlist management while providing clear separation of responsibilities across three types of users: vendors (who rent stalls and sell goods), market organizers (staff who manage the market operations and vendor relations), and system administrators (staff responsible for maintaining the system, account management, and reporting).

2. System Features

2.1 Market Schedule

The system maintains a Market Schedule that contains all planned market dates for the season. Each entry in the Market Schedule represents a specific market date (a Sunday during the season) and tracks the date, total stall availability, stall availability by vendor category (Food and Artisan), currently booked vendors, pricing for that date, and booking status.

Vendors interact with the Market Schedule to browse available market dates and book stalls. The Market Schedule displays only market dates that fall within the next four weeks from the current date and have not yet passed their booking deadline. Market dates remain bookable until Wednesday at 23:59 (11:59 PM) of the week before that market Sunday. Once the Wednesday deadline passes, that market date is automatically removed from the vendor booking interface.

This means vendors see a rolling window of upcoming market dates they can still book. For example:

- On Wednesday, May 6th at noon, vendors would see: May 10th, May 17th, May 24th, and May 31st
- On Thursday, May 7th, vendors would see: May 17th, May 24th, May 31st, and June 7th (May 10th is not available/not visible to the vendor because the Wednesday booking deadline has passed)

As time progresses, new market dates automatically appear in the Market Schedule as they enter the four-week window, and dates automatically become unavailable/not visible to vendors after their Wednesday booking deadline passes.

Organizers and administrators have full access to view the entire Market Schedule including all market dates for the season, not just those within the current booking window. This allows organizers to plan ahead and configure future market dates before they become visible to vendors, and allows administrators to review historical market data and plan future seasons.

2.2 Market Operations

The Hintonville Farmers Market operates seasonally on Sundays only, from the first Sunday in May through the last Sunday in September. Each market day accommodates a maximum of 30 vendor stalls, with specific capacity limits by vendor category to ensure diverse market offerings. Food vendors are limited to a maximum of 20 stalls per market day. Artisan vendors are limited to a maximum of 10 stalls per market day.

2.3 Stall Pricing Structure

Market organizers configure stall rental pricing at the beginning of each season. The default pricing is \$75 per stall per market day. Vendors pay stall rental fees when they book their market dates, and the system tracks payment status for each booking. Vendors with unpaid stall fees will be restricted from booking additional market dates until their account is current.

2.4 Vendor Account Creation

Vendors can create an account at any time by providing business information, and selecting their vendor category. The business information includes business name, owner name, email address, phone number, and mailing address. The vendor category selection must be either Food or Artisan.

Food vendors sell products intended for human consumption, including fresh produce (fruits and vegetables), meat and poultry products, baked goods (breads, cookies, pastries), preserves and canned goods (jams, jellies, pickles), beverages (juices, teas, coffee, non-alcoholic beverages), and prepared foods ready for immediate consumption (sandwiches, soups, meals).

Artisan vendors sell handcrafted items and artistic works, including jewelry, pottery and ceramics, woodwork and furniture, textiles and clothing, paintings and artwork, candles and soaps, and other handmade crafts.

The vendor category selected during account creation determines which compliance documents are required, which market day stall capacity limits apply, and which products the vendor is permitted to sell at the market. Vendors may only select one category per account and may only sell products that fall within their selected category. If a vendor sells both food products and artisan crafts, they must choose their primary category and may only sell products from that category at the Hintonville Farmers Market. Vendors may not create multiple accounts to circumvent the one category per vendor restriction.

Once a vendor submits their account creation request, the account is immediately created and the vendor can begin using the system. The vendor can immediately browse the Market Schedule to view available market dates. However, the vendor cannot complete any stall bookings until they have entered all required compliance documentation information for their category.

2.5 Compliance Documentation Requirements

Before a vendor can complete any stall booking, they must enter all required compliance documentation information specific to their vendor category. The system does not require vendors to upload document files. Instead, vendors declare that they possess the required documents by entering the document information into the system.

Food vendors must enter information for three required documents: a City of Hintonville Business Licence (including license number and expiration date), proof of liability insurance with minimum \$2 million coverage (including policy number, insurance provider, and expiration date), and a valid Ontario Food Handler Certification (including certification number and expiration date). Artisan vendors must enter information for two required documents: a City of Hintonville Business Licence (including license number and expiration date) and proof of liability insurance with minimum \$2 million coverage (including policy number, insurance provider, and expiration date).

All compliance documents must be valid for the entire market season. The system must verify that all required document information has been entered and that all expiration dates extend beyond the last Sunday in September (the end of the market season). If any required documentation information is missing or if any document will expire before the end of the season, the system must prevent any booking from being completed and display a clear message indicating which document information is needed or which documents will expire during the season.

Vendors are responsible for maintaining physical copies of all compliance documents and must present these documents for inspection at their market stall on any market day if requested by market organizers. The system stores the compliance document information entered by vendors, and this information is visible to market organizers and administrators for reference purposes.

2.6 Vendor Status Dashboard

Every vendor must have access to a vendor status dashboard that displays information about their account and market participation. The vendor status dashboard must show the vendor's business information, including business name, owner name, contact information, and vendor category. The dashboard must display the status of compliance documentation, including license and policy details (where applicable, refer to Section 2.5) and their expiration dates.

The vendor status dashboard must display all upcoming market dates for which the vendor has booked stalls, including the market date, payment status (paid or outstanding balance), and booking confirmation details. The dashboard must show all waitlist positions the vendor currently holds, including the market date, and the vendor's position in the queue.

The vendor status dashboard must display system notifications including alerts when the vendor is next in line on a waitlist and a stall has become available for them to book, and confirmation messages when bookings are completed or cancelled. The dashboard must show payment history including all completed payments with dates and amounts, refunds received, and any outstanding balances with due dates. The dashboard must provide access to the vendor's complete transaction history showing all stall bookings, cancellations, and waitlist actions with timestamps.

2.7 Stall Booking

Once a vendor has created an account and entered all required compliance documentation information, they can book stalls for individual market days by browsing the Market Schedule. The Market Schedule displays only those market dates that are currently within the booking window (four weeks in advance until Wednesday before the market Sunday).

Vendors can browse the Market Schedule to view available market dates, see stall availability by vendor category for each date, and book stalls for multiple market days simultaneously in a single transaction.

When booking stalls from the Market Schedule, the system verifies that the vendor has entered all required compliance documentation information for their category. Upon successful booking, a confirmation message is displayed and the booking(s) appear on the vendor's status dashboard. The vendor is booking a stall for each selected market date but is not assigned specific stall locations. Stall locations are managed by market organizers on the day of the market and are not part of this system.

2.8 Stall Cancellation and Refund Policy

Vendors may cancel previously booked market day stalls **up to Wednesday at 12:00 PM before the scheduled Sunday market day**. Cancellations are not permitted after this deadline.

When a vendor cancels a stall booking before the cancellation deadline:

- The system immediately releases the stall and updates stall availability for the affected market date.
- If the released stall creates availability for the vendor's category, the system triggers the appropriate waitlist process for that week and category.
- Vendors receive an automatic refund for the fees paid for cancelled market days. The system records the refund transaction and updates the vendor's payment history.

Cancellations made **after the Wednesday 12:00 PM deadline** are not allowed and do not result in credits or refunds.

All cancellations, refunds, and resulting waitlist actions must be recorded in the system's audit log and reflected in the vendor's transaction history and status dashboard.

2.9 Waitlist Management

When a vendor attempts to book a stall for a market day shown in the Market Schedule and no stalls are available for their vendor category, the system must offer the vendor the option to join a waitlist for that market date. The system maintains waitlists organized by vendor category and by week. Since the booking window shows approximately four weeks of market dates at any

given time, the system maintains a maximum of eight waitlists: one for Food vendors for each of the four visible weeks, and one for Artisan vendors for each of the four visible weeks.

Each waitlist is identified by the vendor category and the week containing the market date. For example, if a food vendor attempts to book a stall for Sunday June 8th when all 20 food vendor slots are filled, the system presents the option to join the waitlist for Food vendors for the week of June 8th. The vendor may accept the waitlist position or decline. If the vendor accepts, they are added to the end of the queue for that category and week in first-come, first-served (FIFO) order, and the waitlist position immediately appears on their vendor status dashboard showing their position in the queue.

When a cancellation creates an available stall for a category on a specific date, the system identifies the appropriate waitlist (based on the vendor category and the week containing that date) and notifies the first vendor in that queue. The notification is posted to that vendor's status dashboard informing them that a stall has become available for the specific market date, and that they have 24 hours to accept and complete the booking. If the vendor accepts and completes the booking within 24 hours, they are removed from the waitlist and the stall is confirmed. If the vendor declines or does not respond within 24 hours, they are removed from that waitlist and the system automatically notifies the next vendor in the queue. Vendors may view all waitlists they have joined at any time on their status dashboard, including their position in each queue. Vendors may voluntarily remove themselves from any waitlist at any time. Vendors may join waitlists for multiple weeks simultaneously (for example, joining both the Food vendor waitlist for the week of June 8th and the Food vendor waitlist for the week of June 15th).

As the booking window rolls forward and market dates move outside the four week window, the corresponding waitlists for those weeks are archived. New waitlists are created automatically as new market dates enter the four week booking window.

2.10 Vendor Capabilities

Vendors must be able to perform several key functions. They must be able to create a vendor account by providing business information and selecting a vendor category. They must be able to enter and manage compliance documentation information including license numbers, policy numbers, certification numbers, and expiration dates for all required documents. They must be able to view their vendor status dashboard showing all account information, bookings, waitlists, notifications, and payment history. They must be able to browse the Market Schedule to see available market dates within the current booking window. They must be able to view stall availability by vendor category for each market date in the Market Schedule. They must be able to book stalls for specific market days for their category. They must be able to join waitlists when their vendor category is at capacity for desired market dates. They must be able to view their position in all waitlist queues they have joined. They must be able to manage their booking schedule including active stall bookings and waitlist positions. They must be able to cancel stall bookings subject to the market's cancellation policy.

Each vendor profile must store contact information, vendor category, compliance documentation information with expiration dates, booking history, waitlist history, payment records, and notification history.

2.11 Market Organizer Capabilities

Market organizers are responsible for managing market operations and supporting vendor services. They must be able to add market dates to the Market Schedule following the seasonal schedule (Sundays only, from the first Sunday in May through the last Sunday in September). They must be able to configure each market date in the Market Schedule including setting stall availability and pricing. They must be able to view vendor compliance documentation information to verify that vendors have declared their required documents and to reference document numbers and expiration dates. They must be able to create vendor accounts on behalf of walk-in or phone applicants. They must be able to book stalls on behalf of vendors for any market date currently visible in the Market Schedule, respecting all capacity limits and booking windows. They must be able to cancel vendor stall bookings when requested. They must be able to remove vendors from waitlists upon request. They must be able to post system wide announcements or individual vendor notifications to vendor status dashboards regarding market policies and operational updates.

When viewing vendor compliance documentation information, organizers can see which documents have been declared, document numbers, insurance providers, and expiration dates. The system must display clear visual indicators of documentation status (all information entered and valid for entire season, expiring before end of season, expired, or missing) for each vendor.

Organizers have the authority to book stalls on behalf of vendors and can manage waitlist positions, including removing a vendor from a waitlist at the vendor's request. When a stall becomes available due to cancellation, the system automatically identifies the appropriate waitlist and notifies the first vendor in that queue through their status dashboard.

Organizers have full access to view the entire Market Schedule including all market dates and bookings for the season, not just those within the current four week booking window. This allows organizers to plan ahead and configure future market dates before they become visible to vendors. Organizers do not assign specific stall locations through the system. Stall location assignments are managed by organizers on-site on the day of each market based on vendor setup needs and market layout considerations.

2.12 System Administrator Capabilities

System administrators provide high level oversight, financial management, and compliance monitoring. They must be able to create market organizer accounts and vendor accounts. They must be able to configure system wide market parameters including maximum vendor counts by category. They must be able to add market dates to the Market Schedule and configure seasonal market schedules. They must be able to generate reports on vendor participation by market date, showing which vendors have booked stalls for which event dates and which

vendors are on waitlists for specific weeks. They must be able to generate financial reports showing stall booking revenue and vendors with outstanding payment balances. They must be able to monitor waitlist status across all active waitlists (up to eight waitlists at any given time). They must be able to view all vendor compliance documentation information system wide. They must be able to post system wide notifications that appear on all vendor status dashboards.

Administrators have full access to view the entire Market Schedule including all market dates for the season, not just those within the current four week booking window. This allows administrators to review historical market data and plan future seasons.

Reports must be exportable in CSV format. Administrators must ensure access controls are enforced, maintain system integrity, and preserve detailed audit logs of all stall bookings, waitlist actions, payments, and administrative actions. All administrative operations must be fault tolerant, succeeding fully or failing gracefully without leaving the system in an inconsistent state.

2.13 Data Consistency and Audit Trail

Across all roles, HintonMarket must maintain accurate and consistent data. Every transaction must be recorded in an audit log to support accountability, including account creation, stall bookings and cancellations, refunds, waitlist joins and removals, payments, compliance documentation entries, and administrative actions. The system must enforce booking policies consistently, prevent data duplication, maintain accurate FIFO queue ordering for all waitlists, and maintain a usable, professional interface that distinguishes clearly between vendors, organizers, and administrators.

3. Technical Specifications

3.1 User Interface

The HintonMarket system must provide a graphical user interface built with Qt widgets. Navigation of features should be accomplished through menus, dialog boxes, or context menus, and the interface must clearly distinguish between vendor, organizer, and administrator roles. The interface should remain consistent across all roles, meaning that navigation patterns, terminology, and the placement of common controls must not change between roles. This ensures an intuitive, professional experience for all users.

Common operations, such as browsing the Market Schedule, booking stalls, or reviewing vendor accounts, must be straightforward to complete. The vendor status dashboard must be prominently accessible from the vendor interface, allowing vendors to quickly check notifications, bookings, and waitlist positions. The Market Schedule interface must clearly display which market dates are currently available for booking and provide real-time updates on stall availability by vendor category.

The HintonMarket desktop application interface shall comply with Ontario's Accessibility for Ontarians with Disabilities Act (AODA).

3.2 Data Storage

Persistent storage must be provided by SQLite. The Market Schedule data, vendor accounts, compliance documentation information, stall bookings, waitlist queues, transaction histories, and notification records must be stored in this database and loaded whenever the system is launched. All modifications must be written immediately to persistent storage to prevent data loss. The database must support efficient searching and sorting operations, and duplication of records must be avoided to preserve data integrity. The system must ship with at least twenty vendor accounts of various category types and ten market dates in the Market Schedule preconfigured.

All dates in the system must be stored and displayed following the ISO 8601 format: YYYY-MM-DD standard. This includes market dates in the Market Schedule, compliance document expiration dates, booking dates, payment dates, and all timestamp information in audit logs and transaction histories.

All vendor personal and business information must be stored and managed in compliance with Canada's federal privacy legislation, the Personal Information Protection and Electronic Documents Act (PIPEDA), secure access controls, and documented data retention policies.

3.3 Data Management and Reporting

The system must support farmers market operations such as creating and managing market dates in the Market Schedule with configurable stall pricing, creating vendor accounts with business information validation (including valid email format requirements), processing stall bookings and payments, maintaining vendor compliance documentation information, managing waitlist queues with FIFO ordering, and managing system notifications displayed on vendor status dashboards.

The Market Schedule must efficiently track stall availability for each market date by vendor category, automatically update as bookings are made or cancelled, and properly enforce the four week booking window visibility rules for vendor access while providing full access to organizers and administrators. The waitlist system must maintain up to eight active queues at any given time (two vendor categories multiplied by four weeks in the booking window), properly enforce FIFO ordering within each queue, and automatically archive waitlists as market dates move outside the booking window.

Each vendor record must include compliance documentation information with document numbers and expiration dates, vendor category classification, contact details, notification history, and all information displayed on the vendor status dashboard. Vendor searches must return results within two seconds for collections of up to five hundred vendor accounts. Market Schedule queries must return available dates and stall availability information within two seconds for typical seasonal schedules.

Vendor records must include business contact details, compliance documentation status with expiration tracking, payment history, notification history, and a full transaction history showing all stall bookings, cancellations, and waitlist actions.

The system must provide administrators with reports that offer meaningful insights into market operations. At minimum, reports listing confirmed vendors by market date (showing vendor names and categories) and financial reports showing stall booking revenue and any outstanding payment balances must be generated. Reports must be exportable in CSV format and must be produced in a timely and accurate manner. Every stall booking, payment, compliance documentation entry, waitlist action, notification posting, and administrative action must be logged to create a reliable audit trail with timestamps and user identification.

3.4 Data Integrity and Fault Tolerance

The system must maintain 99% uptime during market operating hours and administrative business hours. Database transactions must be atomic, meaning that complex operations involving multiple database changes (such as processing a vendor stall booking, which updates both the Market Schedule stall availability and the vendor's booking record) must either complete entirely or fail entirely—no partial updates that leave the system in an inconsistent state. Session timeouts must occur after 30 minutes of user inactivity for security purposes.

3.5 Delivery and Deployment

The HintonMarket system must be delivered as a turnkey package. Users must be able to install, build, and launch the system with a single command each inside the provided Ubuntu virtual machine. Clear, step by step installation instructions must be included with the submission to reduce TA workload. Any submission that cannot be installed, built, and launched in this way will receive a grade of zero.

4. Individual Assignments

All assignments must be completed individually. Every assignment must be submitted as a PDF document. Documents must be typed, legible, and professional, including a cover page, page numbers, table of contents, and section numbering. All UML diagrams must be created using professional drawing tools. Hand drawn diagrams will not be accepted.

Assignment #1: Requirements Analysis

The work for this assignment will consist of a document specifying:

- The functional and non-functional requirements for the entire HintonMarket project
- UML use case diagram of the entire system and detailed table-based use case descriptions for a selected subset of the project features

Assignment #2: System Design

The work for this assignment will consist of a document specifying:

- UML class diagrams and UML sequence diagrams for a selected subset of the HintonMarket project

5. Project Deliverables

Registered teams must complete all project deliverables by the specified deadline. Coding deliverables must be submitted as a single tar or zip file containing all source code, data files, configuration scripts, and complete installation, build, and launch instructions.

Deliverable #1: Subsystem Decomposition and Initial Implementation

The work submitted for this deliverable will consist of documentation describing the subsystem decomposition of your system, along with the implementation of selected feature(s) demonstrating their application.

Deliverable #2: Complete System Integration with Database

The work submitted for this deliverable will consist of the final system decomposition diagram in PDF format, the implementation of selected feature(s), comprehensive testing documentation, and a user manual.